

AI and LLM Solutions

ctrlaltcrew — Applied AI Services Reference

This document provides a detailed reference on ctrlaltcrew's AI and LLM-related capabilities. It covers what each type of AI solution is, what problem it solves, example use cases, when a client should consider it, and what information is needed before starting.

1. Generative AI Applications

What It Is

Applications that use a large language model (LLM) to generate text, summaries, drafts, or other content on demand, based on user input or existing data.

What Problem It Solves

Manual content creation (drafting, summarizing, rewriting) is slow and repetitive. Generative AI applications let a client automate first-draft content generation while a human reviews or edits the output.

Example Use Cases

- Automated report or summary generation from raw data
- Marketing copy or product description drafting tools
- Internal tools that summarize long documents or meeting notes

When a Client Should Consider This

- The client has a recurring content-generation task that currently takes significant manual time.
- A human reviewer will remain in the loop to check output quality, especially for external-facing content.

Information Required Before Starting

- Examples of the desired output format and tone
- Source data or documents the generation should draw from
- Where in the client's workflow the tool should sit (standalone tool vs. embedded feature)

2. LLM-Powered Applications (General)

What It Is

Any application where a large language model is embedded as a core feature — for example, text classification, extraction, summarization, or natural-language interfaces layered on top of existing business logic.

What Problem It Solves

Businesses often have data or workflows that would benefit from natural-language understanding but do not have this capability built in. LLM-powered features add that capability to an existing or new application.

Example Use Cases

- Automatically categorizing incoming support tickets or emails
- Extracting structured data from unstructured text (e.g., invoices, forms)
- Natural-language search over structured business data

When a Client Should Consider This

- The client has an existing application that could benefit from a natural-language feature without becoming a full chatbot or agent project.

Information Required Before Starting

- Sample data representative of what the LLM feature will process
- The exact task the feature should perform (classify, extract, summarize, etc.)
- Accuracy expectations and how errors should be handled

3. RAG Systems (Retrieval-Augmented Generation)

What It Is

A RAG system retrieves relevant information from a client's own documents or data (stored as searchable embeddings in a vector database) and provides it to the LLM as context, so the AI can answer questions grounded in that specific content instead of relying only on its general training knowledge.

What Problem It Solves

General-purpose LLMs do not know a client's private business information and can produce incorrect or fabricated answers about it. RAG solves this by grounding responses in the client's actual documents, with the ability to cite sources.

Example Use Cases

- A chatbot that answers questions using a company's product manuals or policy documents
- An internal assistant that searches years of internal documentation
- A customer support assistant that answers from help-center articles

When a Client Should Consider This

- The client has a meaningful body of existing documents, FAQs, or structured content (this is typically the deciding factor between a simple chatbot and a RAG system).
- Accuracy and traceability of answers (referencing the source document) matters for the use case.

Information Required Before Starting

- The documents or content to be used as the knowledge source, in a readable format (PDF, Word, spreadsheets, web pages, etc.)
- How frequently that content changes, to plan update frequency
- Whether answers need to cite specific source documents
- Any content that should be excluded from retrieval (e.g., confidential sections)

4. AI Chatbots (General Conversational Assistants)

What It Is

Conversational interfaces powered by an LLM that can hold a dialogue with a user, answer questions, and guide them toward an outcome such as booking, purchasing, or getting support.

What Problem It Solves

Businesses often cannot staff live chat around the clock, and simple rule-based chatbots feel rigid. An LLM-powered chatbot can hold a more natural conversation and handle a wider range of phrasing.

Example Use Cases

- 24/7 first-line customer support
- Lead qualification on a website
- FAQ answering without requiring a RAG pipeline, when questions are general rather than document-specific

When a Client Should Consider This

- The client wants conversational, always-available interaction without necessarily grounding every answer in private documents (if document grounding is required, this is combined with RAG — see the RAG-Based Knowledge Assistants service).

Information Required Before Starting

- The chatbot's intended scope (what it should and should not answer)
- Escalation process for questions the bot cannot handle
- Channel(s) where the bot should be deployed (website, WhatsApp, etc.)

5. Document Q&A

What It Is

A focused application of RAG that allows a user to upload or reference a set of documents and ask natural-language questions about their content, receiving answers with references back to the source material.

What Problem It Solves

Reading through long documents to find a specific answer is time-consuming. Document Q&A lets users ask directly and get a grounded answer with a pointer to the relevant section.

Example Use Cases

- Searching long contracts or policy documents for specific clauses
- Answering questions from technical manuals or specifications
- Internal research tools for teams working with large document sets

When a Client Should Consider This

- The client's core need is searching and querying a defined document set, rather than a broader ongoing conversational assistant.

Information Required Before Starting

- The document set to be searchable
- Expected question types, to validate retrieval quality during testing
- Access control requirements if different users should see different documents

6. AI Assistants (Task-Oriented)

What It Is

AI assistants that combine conversation with the ability to take limited actions, such as looking up a record, checking availability, or submitting a form on the user's behalf.

What Problem It Solves

A pure Q&A chatbot can inform a user, but many workflows require the assistant to actually do something (check status, create a request). Task-oriented assistants close that gap.

Example Use Cases

- An assistant that checks order status by looking up a database or API
- A booking assistant that checks availability and creates a reservation
- An internal assistant that can create a support ticket on request

When a Client Should Consider This

- The client wants the assistant to perform a small number of well-defined actions, not an open-ended multi-step workflow (for that, see Agentic AI / Workflow Automation).

Information Required Before Starting

- The specific actions the assistant should be able to perform
- Access to the systems/APIs required to perform those actions
- Rules for when the assistant must hand off to a human

7. Agentic AI / Workflow Automation

What It Is

AI agents that plan and execute multi-step tasks autonomously, using a defined set of tools (APIs, databases, file systems, or other services) to complete a workflow with minimal human input, often across more than one system.

What Problem It Solves

Some business processes involve multiple steps across multiple systems (e.g., read an incoming request, look up related data, update a record, send a notification). Agentic AI automates the full sequence rather than a single response.

Example Use Cases

- An agent that processes incoming leads: enriches data, scores them, and updates a CRM
- An agent that compiles a weekly report by pulling data from several sources

- An agent that triages and drafts responses to a shared inbox for human approval

When a Client Should Consider This

- The client has a repetitive multi-step process that follows clear rules and currently consumes significant staff time.
- The client is comfortable defining clear boundaries and, where appropriate, human-approval checkpoints for actions the agent takes.

Information Required Before Starting

- A step-by-step description of the current manual process
- Access details for each system the agent needs to interact with
- Which actions require human approval versus which can be fully automated
- Error-handling expectations (what should happen if a step fails)

8. Tool-Using AI Systems

What It Is

A technical pattern underlying agentic AI, where the LLM is given a defined set of 'tools' (functions it can call, such as a search function, a calculator, or an API request) and decides when to use them while responding to a user or completing a task.

What Problem It Solves

LLMs alone cannot access live data, perform precise calculations reliably, or take actions in external systems. Tool use extends the LLM's capability by letting it call out to purpose-built functions when needed.

Example Use Cases

- A chatbot that calls a live pricing API before answering a pricing question
- An assistant that calls a calendar API to check real availability
- A research assistant that calls a search tool to find current information

When a Client Should Consider This

- This applies to most AI assistant and agentic automation projects; clients do not usually need to request 'tool use' separately, since it is a building block, not a standalone service.

Information Required Before Starting

- The specific external systems or data sources the assistant should be able to query
- Authentication/access credentials for those systems (handled securely per project)

9. Knowledge-Base Assistants

What It Is

A packaged combination of RAG and chatbot interface, purpose-built to serve as a searchable front end for a client's internal or customer-facing knowledge base — closely related to the RAG-Based Knowledge Assistants service and often the most requested AI solution.

What Problem It Solves

Knowledge scattered across documents, wikis, and folders is hard to search effectively. A knowledge-base assistant centralizes access through natural-language search.

Example Use Cases

- Internal IT/HR help-desk assistant
- Customer-facing product knowledge assistant (similar in spirit to this very RAG demo)
- Onboarding assistant for new employees

When a Client Should Consider This

- The client already has a body of reference content and wants a conversational way for employees or customers to access it, rather than manual searching through folders or a static FAQ page.

Information Required Before Starting

- The full set of source documents/content
- Intended audience (internal staff vs. external customers), which affects access control
- Update frequency of the underlying content

10. General Notes on Starting an AI Project

Across all AI/LLM services above, three factors most strongly influence project scope and cost: the amount and quality of source content or data available, the number of systems the AI needs to integrate with, and whether the AI only answers questions versus also takes actions. Providing clear examples of expected inputs and outputs during the initial consultation significantly speeds up estimation and reduces revisions later in the project.